

Shahid Kamal

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EDUCATION

Northeastern University

Boston, MA

MS in Electrical and Computer Engineering (Computer Vision and Machine Learning)

Expected Dec 2026

Relevant Coursework: Machine Learning Operations, Machine Learning, Deep Learning, Advanced Computer Vision, Reinforcement Learning, Algorithms, Robotics Sensing and Navigation

Aligarh Muslim University

India

BE in Electrical Engineering

2019-2023

EXPERIENCE

Checkit Analytics - AI Engineer (California, United States)

Jan 2026 – Aug 2026

- Built LLM-based RAG pipelines for grounded financial question answering and analytics
- Fine-tuned and deployed Qwen3-4B as the in-house backbone LLM, eliminating external API costs
- Built Rumor Check, a misinformation detection app leveraging LLM-as-Judge evaluation, and deployed it on AWS

Centre of Advanced Research in Electrified Transportation - Research Intern (Aligarh, India)

Dec 2022 - Feb 2023

- EV battery systems suffered from inefficiencies, leading to high energy loss during charging
- Optimized EV Battery Management System using Machine Learning, improving charging efficiency by 7%
- Used Random Forrest and LSTMs, reducing energy loss during charging by 10%

Indian Institute of Information Technology - Summer Research Fellow (Allahabad, India)

June 2022 - July 2022

- Worked with the research team on developing a dynamic routing mechanism in Capsule Neural Networks
- Researched different Routing mechanisms in CNNs and presented findings in the seminar
- Developed a hybrid architecture inception_efficientcaps for dynamic routing in CNN Networks

PROJECTS

Lab Lens – Healthcare AI and MLOps Project

Sep 2025 - Dec 2025

- Built an end-to-end healthcare AI system for medical report summarization, diagnostic image analysis, and risk prediction
- Implemented a RAG Q&A pipeline enabling semantic search and patient-friendly insights from complex medical reports
- Designed MLOps workflows with model evaluation, monitoring, and bias checks for reliable and scalable deployment

Contrast Enhanced Emotion Recognition using Deep Learning

Jan 2025 - April 2025

- Built a facial emotion recognition system classifying 7 expressions on the FER2013 dataset using custom CNN and resnet-18
- Implemented CLAHE contrast enhancement pipeline to improve model robustness on low contrast images facial images
- Evaluated baseline vs contrast enhanced preprocessing strategies, achieving 58.19% test accuracy with a custom CNN

Detection of COVID-19 from Chest X-ray images using Deep Learning Techniques

Aug 2022- May 2023

- Designed a custom CNN achieving 97.2% accuracy in COVID-19 classification using deep learning
- Implemented model compression and acceleration techniques to enable deployment on resource-constrained hardware
- Evaluated and validated performance on real-world CXR datasets, achieving clinical-level detection metrics

SKILLS

Programming Languages: Python, MATLAB, C, C++, SQL

Machine Learning frameworks: TensorFlow, PyTorch, Keras, Numpy, Matplotlib, Pandas, Scikit-learn,

Robotics: ROS 2, SLAM, Robotics Perception, Sensor Fusion, Kalman Filter, Navigation

Tools: Git, GitHub, AWS, GCP, CI/CD Integration, Docker, Kubernetes

PUBLICATIONS

- U. Farooq, S. Kamal and M. Sarfraz, "[Deep Learning Model for Edge Devices for COVID-19 Detection from CXR Images](#)," 2024 Tenth International Conference on Bio Signals, Images, and Instrumentation (ICBSII), Chennai, India, 2024, pp. 1-6
- Shahid Kamal, Dr. Mohammad Sarfraz, Sumaiya Fatma, Ifat Al Fatma, Shagufta Parween, Muhammad Al Maathidi, "[Comparative Analysis of deep learning techniques for fast detection of COVID-19 using CXR images](#)", Proceedings of International Conference on Advances in Computational Intelligence and its Applications, 2023, **Taylor and Francis, CRC Press**,